What you do is first-rate: MRI articulation movies made accessible*

Shigeto Kawahara

May 27, 2026

Abstract

This paper reports a collaborative team effort to create MRI-based articulation movies that are accessible for children with hearing difficulties. While the existing MRI articulation movies are useful for college students and grown-up researchers, they can be scary, off-putting and overwhelming for children. My collaborators and I synched the existing MRI videos with an accessible picture. Not only does this succeed in making the movies generally more accessible, it also has the advantage of being able to show very clearly the teeth, the three larynx cartilages, and the distinction between hard palate and soft palate. Our team also came up with a means to explain manner of articulation in an accessible way, by using anthropomorphized molecule dolls. Our final product is available at YouTube `https://www.youtube.com/watch?v=0LuMVBEr5Y0`. This project shows that phonetic research can make a real difference. This paper is dedicated to my PhD advisor, John Kingston.

Dedication

I knew next to nothing about phonetics when I started my PhD study at UMass, Amherst. I only had very basic knowledge—only to the extent that I could do phonological analyses. In fact, I chose the UMass Linguistics department to become a formal phonologist, but John converted me to a laboratory phonologist/phonetician. I took John's intro to phonetics class during the fall

*I am grateful to my collaborators: Yoichi Kitayama (a professional singer; Gospellers), Kumi Makimura (a manga artist) and Kenji Nomura (a medical illustrator), without whom this project would never have existed. I would also like to express my sincere thanks to Akane Kanai and Emi Nomura who taught us the needs that current Japanese children with hearing difficulties face right now, as well as to all the children who provided very honest feedback about our project.

semester of my first year, where he taught me how we actually produce speech: the physiological phase of "the speech chain" (Denes & Pinson 2012). I also remember quite vividly John's seminar on the interface between phonetics and phonology, which convinced me that you need to understand phonetics if you want to do any kind of phonology, even if you take the extreme position that phonetics and phonology are completely separate grammatical modules.

During my fourth and fifth years at UMass, I was lucky enough to be able to serve as John's RA for his NIH-funded project addressing whether the object of speech perception is articulatory gestures or auditory qualities—we also addressed at which point during the speech perception process linguistic knowledge kicks in (Kingston 2005a). Throughout these years, John, I think, treated me as a genuine collaborator rather than a student assistant.

Almost 20 years have now passed since I graduated, and I would like to talk about a project that I have been working on recently—a project that would not have been possible if John weren't my teacher.

The main story

Please allow me to present the content of this project as a personal narrative from my own point of view.

1 The encounter

I need to start my story during the COVID-19 pandemic time. John and I had a couple of Zoom meetings and discussed this, but I—like many other researchers on the planet at that time—was wondering why I was studying what I was studying. Phonetics was not helping the pandemic situation in any way. I felt, to be honest, useless. I remember that John had similar feelings. But that feeling of mine turned out to open up a new door for me.

In the fall semester of 2021, the class at Keio University was still online, and there was this one student taking my intro to phonetics class. He was a professional singer—Yoichi Kitayama from Gospellers. He quickly realized that learning basic phonetic concepts can be extremely useful for professional singers. For example, he noticed that he was using [θ] for /s/, because in Japanese [s] and [θ] are not contrastive. When I showed an MRI movie of the production of [s] with the alveolar constriction, he thought that he may have a better control of this consonant with this gesture. He belongs to an a cappella group, so it was—and is—essential for him to control the duration of consonants in a matter of milliseconds.

We quickly realized that the collaboration between a phonetician and a professional singer can be fruitful, opening up all sorts of new opportunities. We kept talking, yes, just talking—like

John and I kept talking about what the object of speech perception is. In fact, Yoichi was quite interested in this issue himself as an a cappella singer, because his goal in singing is really to *harmonize* with other members of the group. He had this intuitive idea that he was not aiming for particular articulatory settings, but instead was trying to produce "sounds that would echo with others." Yes, just as John hypothesized that we have auditory—instead of articulatory—targets (Kingston 2007), Yoichi had a very similar feeling. He was very happy and excited to see some actual experimental evidence of this view.[1]

My collaboration with Yoichi continued; and in January 2024, together with Kumi Makimura (a manga illustrator), we ended up publishing a picture book "Utau karada-no fushigi" [A wonderful nature of a singing body] (2024, Kodansha), which illustrates the production phase of our speech chain—the very topic that John taught me for the first time. In that picture book, anthropomorphized air molecules travel through our body—the lung, the larynx and the vocal tract.

2 The realization

A year has passed after the publication of the picture book. In January 2025, I was invited to give an online lecture for school teachers. During the Q&A period, one teacher told me that she was teaching special classes for children with hearing difficulties, and that the children in her class found the picture book to be very useful. That was a very gratifying moment, to say the least.

That night, I was telling that story to my wife, who immediately responded back to me, "that seems like an important next step for your future research. Why don't you ask them more about the children?". So I did.

I had a meeting with two teachers who teach those special classes. They told me that in Japan, children born with hearing difficulties can receive cochlear implants as early as 8 months old. This technological advancement has created new possibilities—children with cochlear implants now have the option to attend regular schools with hearing children, expanding beyond the traditional pathway of specialized schools for the deaf.

But this integration has at the same time created a new challenge: because cochlear implants make hearing difficulties less visible, teachers and classmates may not realize that these children need additional support. That's why specialized speech training remains essential. Our picture book was being used for that training.

They continued that one of the children reported that when they saw the scene in the picture

[1] I also taught Yoichi how much of our speech articulation is controlled, a topic for which John's paper is extremely influential (Kingston & Diehl 1994). We also discussed how Yoichi—and other singers—are dealing with F0 perturbation due to voiced/voiceless obstruents during singing, as it can conflict with the song's melody (cf. Kingston 2005b). We also talked about how singers may manipulate several articulatory landmarks, architectural notions that are crucial in Articulatory Phonology (Browman & Goldstein 1986; Kingston & Cohen 1992)...and so on. This list can be quite long.

book where one air molecule travels to the oral cavity while another travels to the nasal cavity, their articulation was generally nasalized—“I see! The air molecules were mistakenly traveling through my nose!” Apparently, the child felt like they were “speaking” through their nose because they could feel the vibration of the nose, but that habit made their speech over-nasalized throughout

Another student, the teachers continued, told them that they would like to see *a movie*, not just pictures. I innocently told the teachers that there are many MRI videos that show the articulation of all the sounds in Japanese.

The teachers knew, and immediately responded, “You know what? MRI videos are very scary for children”.

I, honestly, felt stunned. I didn’t realize that MRI movies can be terrifying. It was an arrogant aspect of myself as a researcher—in order to understand phonetics, you need to be able to understand MRI videos; that is one of the basic skills as a phonetician. But that sort of academic logic means nothing for children who are struggling. I felt ashamed for a while, but I promised the teachers that I would find a way.

3 The serendipity

I made a promise, but I found myself at a loss. I don’t think I can create accessible MRI articulation videos from scratch? I immediately texted Yoichi. He immediately expressed interest and called for a meeting. He suggested that we don’t have to create MRI articulation movies from scratch—there must be a kind of technology that, for example, V-tubers are using.

And here was another miracle—Kumi Makimura, the manga illustrator (our collaborator on the picture book), knew somebody who was a medical illustrator, Kenji Nomura (TOKIA), whose job was to create “medical movies.” And we actually met him a few months ago at a public event related to that picture book. We immediately got in touch, and he said yes. It is possible to synch an accessible picture with the existing movies.

So this is how we formed our team: a phonetician (myself), a singer (Yoichi), a manga illustrator (Kumi) and a medical illustrator (Kenji). Now that I’ve written this out, it reminds me of John’s team back then—there were several students with various strengths, which sometimes resulted in an extensive cross-linguistic comparison (e.g. Kingston et al. 2009). Not only did John teach me phonetics, he taught me how to collaborate.

4 The challenge

Now let's consider Figure 1. What's on the left is what we—phoneticians—are already familiar with.[2] But it can be terrifying, when we put ourselves in children's shoes. It's dark and grey. The spine portion can be overwhelming. And of course, unless you know what you're seeing, you don't know which part corresponds to what—the tongue, the lips, the palate, etc...And the teeth are not even shown! Neither is the distinction between the hard palate and soft palate. The three larynx cartilages (cricoid, thyroid and arytenoid) are barely visible, if at all.

What's provided on the right is what Kumi and Kenji drew based on the original image. Each articulatory organ is more clearly shown, including the distinction between the hard palate and the soft palate. The cricoid, thyroid and arytenoid cartilages are now very clearly visible. And the spine is gone, and overall, it is much more pleasant to observe.

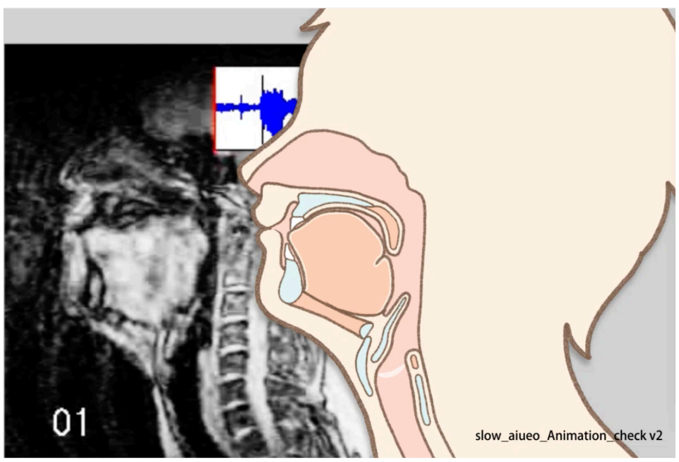

Figure 1: A comparison between the original MRI-image and the renewed image.

5 The delivery

So we created the MRI movies for practically all the sounds in Japanese and delivered them to the children who requested them. I think we did an OK job. The children were pleased to see the edited MRI movies.

But they—and their parents—also requested more. The movies are great, but they make little sense without my explanations. So if they were just provided with these movies, the movies are,

[2]The MRI footage used in this video is an edited version of the MRI movies (created by Kazuhiro Isomura) included in the supplementary data of Onsei o oshieru [Teaching Pronunciation] (Japan Foundation Japanese Language Teaching Series 2, Hituzi Syobo, 2009), used with permission. We thank Kazuhiro Isomura and Hituzi Syobo for granting permission.

well, practically useless. And the children and the parents also wanted to see the airflow. In our parlance, the place of articulation can be easily seen, but the manner of articulation was not "really there."

So here was a new homework assignment for us, but I was at a loss again—how do I illustrate manners of articulation in an accessible fashion?

6 Kawahara-sensei, you should act

The team immediately had a meeting after the delivery. And this is what's miraculous about the interdisciplinary team. Kenji came up with the idea that we can attach "an air molecule doll" at the top of a pointer, and I can "act". Kumi immediately started drawing pictures of molecules illustrating different air flow states. Figure 2 is me acting on the oral pressure build-up behind the stop closure.

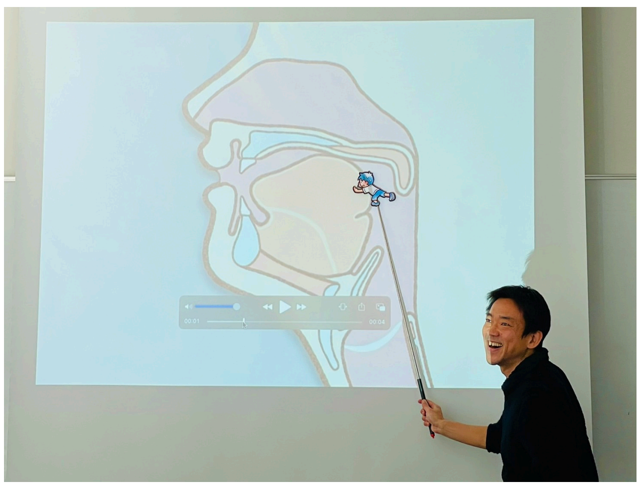

Figure 2: Me acting out—representing the oral pressure buildup behind the stop closure.

Figure 3 shows different dolls representing different air flows. I believe that representing different manners of articulation in human speech in this way is novel—and promising.

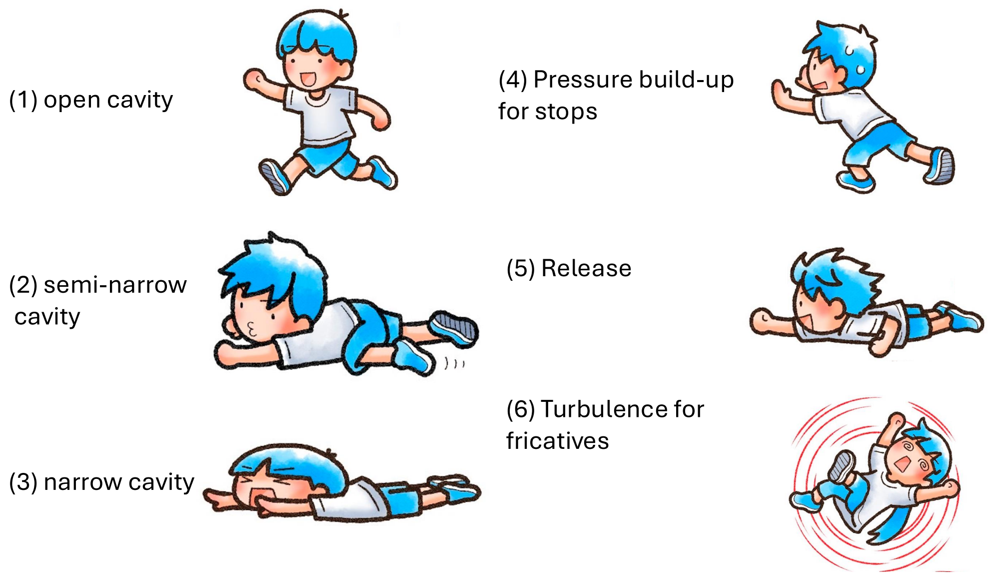

Figure 3: Different air molecule characters representing different manners of articulation. Each character's posture and movement visually captures the aerodynamic characteristics: (1) open cavity for open vowels, (2) semi-narrow cavity for mid vowels, (3) narrow cavity for close vowels and fricatives, (4) pressure build-up for stops, (5) release burst, and (6) turbulence for fricatives.

7 The product

Now we have the edited MRI movies which now clearly illustrate the place of articulation. And I can act as molecules traveling through the vocal tract to explain the manner of articulation. Voicing is hard to explain with the MRI movies, but let's set that aside.[3] The next challenge was—how do we deliver our product to those who need it?

Yoichi, being a professional singer for more than 30 years, had an idea and a capability to make that happen: collaborate with an influential YouTuber. We discussed and tested how to convey basic phonetic concepts to viewers with no prior knowledge. And the final product is available at `https://www.youtube.com/watch?v=0LuMVBEr5Y0`. In that movie, we also made sure to compare sound pairs that are difficult to distinguish, like [su] and [tsu] and [ɕi] and [çi].

And so far we have received only positive comments from various people, not just from children with hearing difficulties but also from language teachers, singers and broadcasters.

8 Looking ahead

Remember our conversation about "phonetics being useless" during the COVID pandemic? I feel quite the contrary now. With this movie, I was able to show how Japanese sounds are produced in the—dare I say—most accessible and explicit ways ever possible.

But we are not done yet—we want to record articulatory gestures of other speakers, like those of singers and broadcasters. We want to make similar accessible videos on articulations of sounds in other languages. We also want to provide coronal views of speech articulations, etc. But for now, this is what I have as of May 2026.

Happy retirement, John.

References

Browman, Catherine & Louis Goldstein. 1986. Towards an articulatory phonology. *Phonology Yearbook* 3. 219–252.

Denes, Peter B. & Elliot T. Pinson. 2012. *The speech chain: The physics and biology of spoken language.* Literary Licensing, LLC.

Kingston, John. 2005a. From ears to categories: New arguments for autonomy. In Sónia Frota, Marina Vigário & Maria João Freitas (eds.), *Prosodies: With special reference to Iberian languages*, 177–222. Berlin: Mouton de Gruyter.

Kingston, John. 2005b. The phonetics of Athabaskan tonogenesis. In Sharon Hargus & Keren Rice (eds.), *Athabaskan prosody*, 137–184. Amsterdam: John Benjamins.

[3] This still remains as a genuine challenge for us.

Kingston, John. 2007. The phonetics-phonology interface. In Paul de Lacy (ed.), *The Cambridge handbook of phonology*, 401–434. Cambridge: Cambridge University Press.
Kingston, John & Avis H. Cohen. 1992. Extending articulatory phonology. *Phonetica* 49. 194–204.
Kingston, John & Randy Diehl. 1994. Phonetic knowledge. *Language* 70. 419–454.
Kingston, John, Shigeto Kawahara, Della Chambless, Daniel Mash & Eve Brenner-Alsop. 2009. Contextual effects on the perception of duration. *Journal of Phonetics* 37(3). 297–320.